# Mathematics for Biomedical Engineering

## L5:

## **Integration**

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October 29th, 2025

- To appreciate that integration is differentiation in reverse;
- To understand the interpretation of the definite integral as the limit of a sum;
- To learn a variety of techniques for obtaining integrals.

1. Integration
   - Interpretation of Integration
   - Calculation of Indefinite Integrals
   - Definite Integrals: Integration as The Limit of a Sum
2. Techniques of Integration
   - Integration by Parts
   - Integration by Substitution
   - Integration Using Partial Fractions
   - Integration of Trigonometric Functions

## Differentiation in reverse

**Integration** is the process of obtaining a function, $f(x)$, from knowledge of its derivative, $\frac{\mathrm{d}f}{\mathrm{d}x}$.

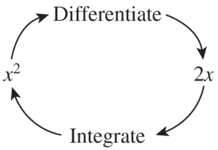



Figure: Integration can be thought of as differentiation in reverse.

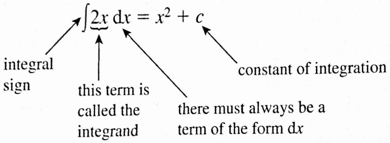



Figure: Nomenclature.

Technically, integrals of this sort are called **indefinite integrals** (the answer must always contain a **constant of integration**).

# Antiderivative

In general we have

$$\int f(x)\mathrm{d}x = F(x) + c,$$

with $c$ the integration constant.

The function $F(x)$ is a differentiable function whose derivative is equal to the original function $f(x)$; i.e.

$$\frac{\mathrm{d}F(x)}{\mathrm{d}x} = f(x).$$

$F(x)$ is called the **antiderivative** of $f(x)$.

The process of solving for antiderivatives is called **antidifferentiation** or **indefinite integration**. Its opposite operation is called **differentiation**.

# Common functions and their indefinite integrals

| Function $f(x)$ | Indefinite integral $\int f(x)\,\mathrm{d}x$ | |
|---|---|---|
| constant, $k$ | $kx + c$ | |
| $x$ | $\frac{x^2}{2} + c$ | |
| $x^2$ | $\frac{x^3}{3} + c$ | |
| $x^n$ | $\frac{x^{n+1}}{n+1} + c$ | $n \neq -1$ |
| $x^{-1} = \frac{1}{x}$ | $\ln\lvert x\rvert + c$ | |
| $\sin x$ | $-\cos x + c$ | |
| $\cos x$ | $\sin x + c$ | |
| $\sin kx$ | $\frac{-\cos kx}{k} + c$ | |
| $\cos kx$ | $\frac{\sin kx}{k} + c$ | |
| $\tan kx$ | $\frac{1}{k}\ln\lvert\sec kx\rvert + c$ | |
| $\sec kx$ | $\frac{1}{k}\ln\lvert\sec kx + \tan kx\rvert + c$ | |

$k$ and $n$ are constants, and all angles are in radiants; $c$ is the constant of integration.

# Common functions and their indefinite integrals

| | | |
|---|---|---|
| $e^x$ | $e^x + c$ | |
| $e^{-x}$ | $-e^{-x} + c$ | |
| $e^{kx}$ | $\dfrac{e^{kx}}{k} + c$ | |
| $\cosh kx$ | $\dfrac{1}{k}\sinh kx + c$ | |
| $\sinh kx$ | $\dfrac{1}{k}\cosh kx + c$ | |
| $\dfrac{1}{x^2 + a^2}$ | $\dfrac{1}{a}\tan^{-1}\dfrac{x}{a} + c$ | $a > 0$ |
| $\dfrac{1}{x^2 - a^2}$ | $\dfrac{1}{2a}\ln\dfrac{x-a}{x+a} + c$ | $\lvert x\rvert > a > 0$ |
| $\dfrac{1}{a^2 - x^2}$ | $\dfrac{1}{2a}\ln\dfrac{a+x}{a-x} + \mathrm{c}$ | $\lvert x\rvert < a$ |
| $\dfrac{1}{\sqrt{x^2 + a^2}}$ | $\sinh^{-1}\left(\dfrac{x}{a}\right) + c$ | $a > 0$ |
| $\dfrac{1}{\sqrt{x^2 - a^2}}$ | $\cosh^{-1}\left(\dfrac{x}{a}\right) + c$ | $x \geq a > 0$ |
| $\dfrac{1}{\sqrt{x^2 + k}}$ | $\ln(x + \sqrt{x^2 + k}) + c$ | |
| $\dfrac{1}{\sqrt{a^2 - x^2}}$ | $\sin^{-1}\left(\dfrac{x}{a}\right) + c$ | $-a \leq x \leq a$ |

$k$, $n$ and $a$ are constants, and all angles are in radiants; $c$ is the constant of integration.

# Basic Rules

The following rules of integration enable us to extend the range of functions that we can integrate.

### Integral of the sum or difference of two functions

The integral of $f(x) \pm g(x)$ is

$$\int (f(x) \pm g(x))\,\mathrm{d}x = \int f(x)\mathrm{d}x \pm \int g(x)\mathrm{d}x \tag{1}$$

### Integral of a function multiplied by a constant

The integral of $k\,f(x)$ is

$$\int k\,f(x)\mathrm{d}x = k \int f(x)\mathrm{d}x \tag{2}$$

# Definite Integrals

They have many applications; e.g. in finding areas bounded by curves, and finding volumes of solids.

The quantity

$$\int_{x=a}^{x=b} f(x)\mathrm{d}x \text{ or } \int_a^b f(x)\mathrm{d}x$$

is called the **definite integral** of $f(x)$ from $a$ to $b$.

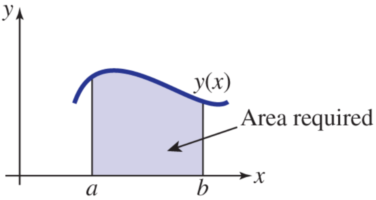
**Important**

If you interchange the limits, you must introduce a minus sign; i.e.

$$\int_a^b f(x)\mathrm{d}x = -\int_b^a f(x)\mathrm{d}x$$

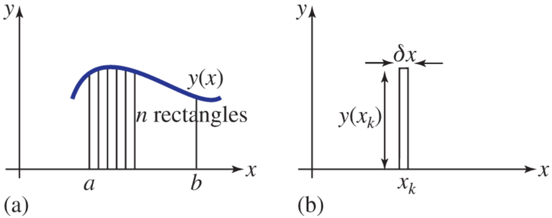
# Integration as a Limit of a Sum

Integration can be also regarded as a process of adding up, or **summation**.
Suppose we are interested in finding the area under a curve $y(x)$ between $a$ and $b$.



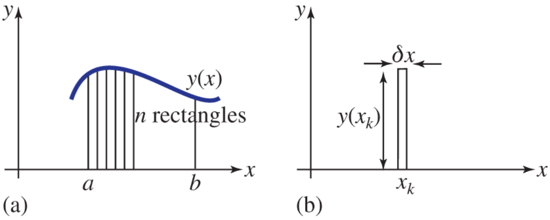

The area can be approximated by dividing it into $n$ rectangles (a). Figure (b) shows the dimensions of a typical rectangle that is located at $x = x_k$.

- The width of each rectangle is $\delta x = \frac{b-a}{n}$;
- The area is $y(x_k)\delta x$;
- The sum of the areas is $\sum_{k=1}^{k=n} y(x_k)\delta x$.



# Integration as a Limit of a Sum

$\sum_{k=1}^{k=n} y(x_k)\delta x$ provides an estimate of the area under the curve.




To improve the estimate we must take a larger number of very thin rectangles.
The exact value of the area is therefore given by

$$\lim_{n\to\infty}\sum_{k=1}^{n} y(x_k)\delta x$$

Note that when $n \to \infty$ we have that $\delta x \to 0$. Hence, the above expression is equivalent to

$$\lim_{\delta x\to 0}\sum_{x=a}^{x=b} y(x)\delta x$$

# Integration as a Limit of a Sum

Indefinite integrals (or antiderivatives) are related to definite integrals through the **fundamental theorem of calculus**.

## Fundamental Theorem of Calculus

It can be shown that

$$\lim_{\delta x \to 0} \sum_{x=a}^{x=b} f(x)\delta x = \int_a^b f(x)\mathrm{d}x = F(b) - F(a)$$

Hence, the definite integral of a function over an interval $[a, b]$ is equal to the difference between the values of an antiderivative, $F(x)$, evaluated at the endpoints of the interval.

`Example:` Find $\int_1^4 x^2 \mathrm{d}x$

# Integration by Parts

This technique is used to integrate the **product of two functions**.
You can think of this as a **product rule for integrals**.

## Integration by Parts Formula

For indefinite integrals:

$$\int u \left(\frac{\mathrm{d}v}{\mathrm{d}x}\right) \mathrm{d}x = uv - \int v \left(\frac{\mathrm{d}u}{\mathrm{d}x}\right) \mathrm{d}x$$

For definite integrals:

$$\int_a^b u \left(\frac{\mathrm{d}v}{\mathrm{d}x}\right) \mathrm{d}x = [uv]_a^b - \int_a^b v \left(\frac{\mathrm{d}u}{\mathrm{d}x}\right) \mathrm{d}x$$

`Example:` Find $\int x \sin x \, \mathrm{d}x$

Sometimes it is necessary to apply the formula twice.
`Example:` Find $\int_0^2 x^2 e^x \, \mathrm{d}x$

# Integration by Substitution

It involves making a substitution to simplify an integral or *make it more amenable to integration*.

It is the **counterpart to the chain rule of differentiation**.

**Key point**

Given the functions $f(x)$ and $g(u)$,

$$\int f(x)\,\mathrm{d}x = \int f(g(u))\,g'(u)\,\mathrm{d}u$$

for indefinite integrals and

$$\int_{g(a)}^{g(b)} f(x)\,\mathrm{d}x = \int_a^b f(g(u))\,g'(u)\,\mathrm{d}u$$

for definite integrals, with $x = g(u)$ in both cases.

The choice of which substitution to make often relies upon experience.

`Example:` Find $\int (3x+5)^6\,\mathrm{d}x$

# Integration by Substitution

Note that when dealing with definite integrals, the limits must be written in terms of the new variable.

`Example:` Find $\int_2^3 t \sin\left(t^2\right) \mathrm{d}t$

## Integrals of the form $\int \frac{f'(x)}{f(x)} \, \mathrm{d}x$

The result is the logarithm of the denominator:

$$\int \frac{f'(x)}{f(x)} \, \mathrm{d}x = \ln |f(x)| \; + \; \text{constant}$$

`Example:` Find $\int \frac{2x}{x^2+8} \, \mathrm{d}x$

# Some Harder Examples

| $f(x)$ contains | make the substitution: | |
|---|---|---|
| $\sqrt{a^2 - x^2}$ | $x = a\sin\theta$ | $\dfrac{dx}{d\theta} = a\cos\theta$ |
| | $x = a\tanh u$ | $\dfrac{dx}{du} = a\,\mathrm{sech}^2\,u$ |
| $\sqrt{a^2 + x^2}$ | $x = a\sinh u$ | $\dfrac{dx}{du} = a\cosh u$ |
| | $x = a\tan\theta$ | $\dfrac{dx}{d\theta} = a\sec^2\theta$ |
| $\sqrt{x^2 - a^2}$ | $x = a\cosh u$ | $\dfrac{dx}{du} = a\sinh u$ |
| | $x = a\sec\theta$ | $\dfrac{dx}{d\theta} = a\sec\theta\tan\theta$ |
| $\sin x$, $\cos x$ | $\sin x = \dfrac{2t}{1+t^2}$, $\cos x = \dfrac{1-t^2}{1+t^2}$, where $t = \tan\dfrac{x}{2}$ | $\dfrac{dx}{dt} = \dfrac{2}{1+t^2}$ |
| $\sin^2 x$, $\cos^2 x$ | $\sin x = \dfrac{t}{\sqrt{1+t^2}}$, $\cos x = \dfrac{1}{\sqrt{1+t^2}}$, where $t = \tan x$ | $\dfrac{dx}{dt} = \dfrac{1}{1+t^2}$ |

# Integration Using Partial Fractions

Integrals of algebraic fractions may be simplified by writing the fraction as a sum of simpler fractions.

Previously described techniques may need to be used to integrate the simpler fractions (see Topic 1).

`Example:` Find $\int \frac{23-x}{(x-5)(x+4)}\,\mathrm{d}x$

# Integration of Trigonometric Functions

We have previously seen that:

$$\int \sin kx \, \mathrm{d}x = -\frac{\cos kx}{k} + \text{ constant}, \qquad \int \cos kx \, \mathrm{d}x = \frac{\sin kx}{k} + \text{ constant}$$

Integrals involving products and powers of trigonometrical functions can be solved by re-writing the integrand in an alternative form using **trigonometrical identities**.

Unfortunately, there is no single approach for tackling these integrals. Sometimes you may need to integrate by parts or by substitution.

`Example:` Find $\int sin^2 x \, \mathrm{d}x$

# What We Learnt Today

- To appreciate that integration is differentiation in reverse;
- To understand the interpretation of the definite integral as the limit of a sum;
- To learn a variety of techniques for obtaining integrals.